



C23-EC-ECII-306

23134

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – THIRD SEMESTER EXAMINATION

PROGRAMMING IN C AND MATLAB

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define the keywords and list any four of them.
2. List any three logical and three bitwise logical operators supported by C.
3. Write the syntax of conditional expression.
4. Differentiate while and do while loops.
5. List any three string manipulation functions.
6. State the use of return statement.
7. Define a structure and write the syntax of struture declaration.
8. List the six preprocessor directives.
9. Give any three major differences between MATLAB and C.
10. State the usage of linspace operator in Matlab.

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1

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PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** (a) Give the syntax of printf() and scanf() functions. 5
(b) List the compound/ shorthand assignment operators. 5
- 12.** Write a program to check whether a given number is Armstrong or not. 10
- 13.** Write a C-program to find largest number in an array. 10
- 14.** Explain different functions used for reading and writing strings with examples. 10
- 15.** (a) Explain passing parameters to function. 5
(b) Differentiate address and dereferencing operators. 5
- 16.** Illustrate a structure with a program to read and print a book database title, author and price as fields. 10
- 17.** (a) State the need for Matlab in solving engineering problems. 5
(b) State the usage of (i) SIMULINK and (ii) GUI. 5
- 18.** Explain the plot commands plot(x,y), ezplot(), fplot(), bar(x,y) and pie(X) in Matlab.

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